Formula 1 Podium Prediction Companion

Milestone 1 · Analytical Report

This report gives the written answers to the three data-engineering questions with their figures, the selected model features and why each is valid at the prediction cutoff (after qualifying, before the race starts), and the known limitations. All numbers and figures are outputs of the team's run-all Kaggle notebook, which contains the full audit, cleaning, modeling, experiments and explainability. Unit of analysis: one driver entering one race; target: podium (positionOrder ≤ 3).

Data used by all three answers. 26,759 race results were reduced to one row per driver per race (26,668 rows; 85 shared-drive groups resolved by a car-based rule) and then to the **modeling cohort** of 24,603 drivers who actually took the start, excluding the Indianapolis 500 of 1950–1960. Non-starters, who can never reach the podium, are left out of every denominator.

1. Data-engineering questions

1.1 Q1. Which circuits convert a front-row start into a podium most often, and does that change after 2014?

Method. Front row = starting-grid positions 1–2, after penalties. Joins: results → races → circuits for the rates; qualifying (one row per driver–race, verified) for the grid vs qualifying comparison. Conversion = podiums ÷ front-row starts per circuit and era, with 95% Wilson intervals, which stay reliable for small samples. A circuit is ranked only with at least 20 front-row starts (about 10 races) in an era, and rankings are compared with Spearman rank correlation. Because 1950–1993 had much lower reliability, the comparison is repeated against 1994–2013 only.

> **Answer.** Front-row conversion rose from **56%** (992 of 1,771 front-row starts, 1950–2013) to **76%** (345 of 454, 2014–2024), and the circuit ranking changed: Spearman −0.18 across the seven circuits ranked in both eras, −0.43 against 1994–2013. Since 2014, Yas Marina, Monaco, Spa and Barcelona convert most often.

Table 1. Top five circuits by front-row conversion in each era (29 circuits ranked before 2014, 10 from 2014).

Rank	1950–2013	Rate (n) [95% CI]	2014–2024	Rate (n) [95% CI]
1	Magny-Cours	80.6% (36) [65–90]	Yas Marina	86.4% (22) [67–95]
2	Sepang	70.0% (30) [52–83]	Monaco	85.0% (20) [64–95]
3	Suzuka	70.0% (50) [56–81]	Spa-Francorchamps	81.8% (22) [61–93]
4	Hungaroring	67.9% (56) [55–79]	Barcelona-Catalunya	81.8% (22) [61–93]
5	Estoril	65.4% (26) [46–81]	Circuit of the Americas	80.0% (20) [58–92]

Table 2. Circuits ranked in both eras: Monaco rose from 6th to 1st, the Hungaroring fell from 1st to last.

Circuit	Starts 1950–2013	Rate	Rank	Starts 2014–2024	Rate	Rank
Circuit de Monaco	120	49.2%	6	20	85.0%	1
Spa-Francorchamps	92	53.3%	4	22	81.8%	2
Barcelona-Catalunya	46	65.2%	2	22	81.8%	2
Silverstone	94	56.4%	3	24	75.0%	4
Red Bull Ring	50	52.0%	5	26	73.1%	5
Monza	128	46.9%	7	22	72.7%	6
Hungaroring	56	67.9%	1	22	59.1%	7

About 8 of the 20 points of overall increase reflect the poor reliability of early decades: against 1994–2013 (64.4%) the rise is about 12 points. The rest is consistent with more reliable cars and with dominant teams locking out the front row (Mercedes 2014–2020, Red Bull 2022–2023); the data cannot separate these two causes.

Table 3. Grid vs qualifying front row: podium conversion (qualifying data exists from 1994).

Era	Front row by grid	Front row by qualifying	Qualified front row, started further back
1994–2013	65.5% (530 starts)	64.9% (530)	22.2% (9 drivers)
2014–2024	76.0% (454 starts)	74.0% (454)	33.3% (27 drivers)

Grid vs qualifying. Drivers who qualified on the front row but started further back, mostly after grid penalties, rose from 9 to 27 and still reached the podium 33% of the time in 2014–2024, about twice the overall podium rate. Qualifying pace carries information beyond the grid slot, which is why grid and qualifying are kept as separate features.

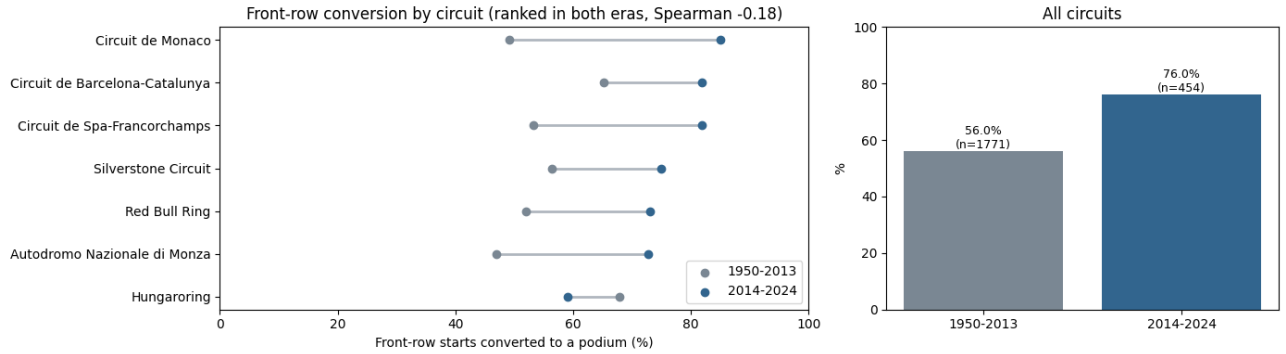


Figure 1. Q1. Left: front-row conversion by circuit for circuits ranked in both eras, 1950–2013 (grey) vs 2014–2024 (blue). Right: conversion over all circuits.

Limitations. Only 6–7 circuits have enough front-row starts in both eras, and post-2014 rates rest on 20–26 starts each, so the intervals overlap: the reshuffle and the overall rise are supported, the exact order of individual circuits is not. A front row is defined as grid 1–2 in every era, although 1950s front rows held three or four cars.

Grain. one row per (raceId, driverId) in the modeling cohort.
Missingness. grid is never missing; the 72 pit-lane starts (grid = 0) are not front-row starts; qualifying positions exist only from 1994 (10,441 of 12,247 cohort rows in 1994–2024), so the grid vs qualifying comparison is restricted to those seasons and nothing is imputed; circuits with fewer than 20 front-row starts are shown but not ranked.
Pre-race cutoff. grid and qualifying position are known before the race; the podium is the outcome being described. This is a descriptive analysis, not a model feature.

1.2 Q2. Do drivers podium more often at a circuit in their home country, after controlling for starting position?

Method. Joins: results → drivers (nationality) and results → races → circuits (country), linked by an explicit nationality-to-country mapping: all 43 nationalities in the cohort are mapped, home is decided by circuit country, dual nationalities count in either country, and an unmapped value stops the notebook so nothing silently becomes “away”. The 72 pit-lane starts are left out because they have no comparable grid slot. Home and away starts are compared only within the same grid band (1, 2, 3, 4–6, 7–10, 11–15, 16–20, 21+) and the same era (1950–1979, 1980–2013, 2014–2024): 24 strata, all with both home and away starts. Era is controlled because home races were 23% of starts in the 1950s and about 3% since 2010. The strata are combined with a weighted difference in podium rate (weights = home starts) and a Mantel–Haenszel odds ratio, each with a 95% interval.

Answer. No. Comparing drivers who started from the same grid band in the same era, home drivers reach the podium 0.8 percentage points *less* often (95% CI −2.0 to +0.5), odds ratio 0.90 (0.76–1.07). The data rule out a home advantage larger than about half a percentage point.

Unadjusted, home drivers reach the podium less often (11.7% of 1,926 home starts vs 13.8% of 22,605 away starts) because they start further back: 14.6% of home starts are from 21st or lower, against 10.9% of away starts. Starting position is a confounder; once it is controlled, the difference disappears.

Table 4. Home vs away podium rate, compared within grid band and era.

Scope	Home starts	Adjusted difference (pp) [95% CI]	Mantel–Haenszel OR [95% CI]
All eras (grid band × era)	1,926	−0.78 [−2.01, +0.46]	0.90 [0.76, 1.07]
1950–1979	883	−0.73 [−2.56, +1.11]	0.91 [0.70, 1.18]
1980–2013	899	−0.66 [−2.44, +1.13]	0.92 [0.71, 1.18]
2014–2024	144	−1.83 [−6.42, +2.76]	0.79 [0.43, 1.46]
Robustness: coarse bands (11+ merged)	1,926	−0.89 [−2.12, +0.35]	0.89 [0.75, 1.06]
Robustness: fine bands (slots 1–10)	1,926	−0.71 [−1.92, +0.50]	0.91 [0.76, 1.08]

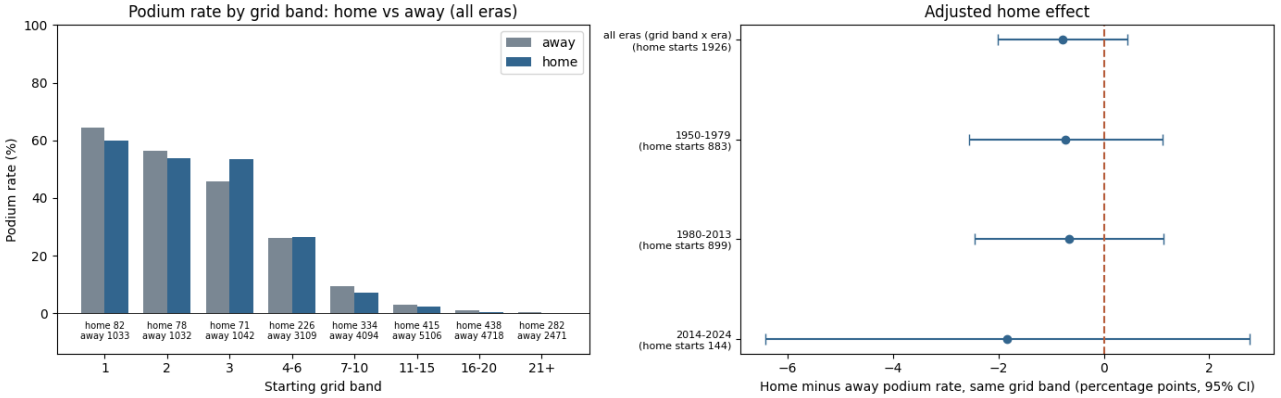


Figure 2. *Q2. Left: podium rate by grid band, home vs away (all eras pooled; counts under each band). Right: adjusted home effect with 95% intervals; every interval crosses zero.*

Limitations. This is an association, not a causal effect. The modern era has only 144 home starts, so its estimate is very uncertain. Rows from the same race are not independent (only three drivers can podium), so the intervals are approximate. Differences the grid band does not capture, such as car quality within a band, are not controlled.

Grain. one row per (raceId, driverId) in the modeling cohort.
Missingness. all 43 cohort nationalities are mapped (an unmapped value stops the notebook); the 72 pit-lane starts are excluded from the grid-band comparison; every one of the 24 strata has both home and away starts.
Pre-race cutoff. driver nationality, circuit country and grid position are known before the race; the podium is the outcome being described.

1.3 Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Method. Joins: results → races (season), constructors (identity) and status (cause). Every status from 2014 onward is mapped to one cause: finished (including "+N Laps"), mechanical (36 statuses such as Engine, Power Unit, Gearbox, Hydraulics, Brakes), accident/damage, driver/rules (disqualified, excluded, illness) or ambiguous (Retired, Puncture, Wheel, Tyre, wings and similar; 88 starts). An unmapped status stops the notebook. The denominator is actual starts, because a mechanical retirement is a start that ended early through a component failure: 4,607 starts (3,255 in 2014–2021, 1,352 in 2022–2024). Rate = mechanical retirements ÷ starts with a 95% Wilson interval; a constructor needs at least 40 starts (about one season with two cars) to be ranked. Ambiguous causes are excluded from the main rate and tested as a sensitivity.

Answer. Mechanical retirements fell from **9.1%** of starts (297 of 3,255) in 2014–2021 to **5.6%** (75 of 1,352) in 2022–2024. The highest rates were Lotus F1, Toro Rosso, McLaren and Renault in 2014–2021, and Alfa Romeo and Alpine in 2022–2024.

Table 5. Highest and lowest mechanical-retirement rates per era (constructors with at least 40 starts).

	2014–2021 (17 ranked)	Rate (starts) [95% CI]	2022–2024 (12 ranked)	Rate (starts) [95% CI]
Highest	Lotus F1	17.3% (75) [10.4–27.4]	Alfa Romeo	12.5% (88) [7.1–21.0]
2nd highest	Toro Rosso	16.1% (242) [12.0–21.3]	Alpine F1 Team	11.0% (136) [6.8–17.4]
3rd highest	McLaren	13.5% (318) [10.2–17.7]	AlphaTauri	6.8% (88) [3.2–14.1]
4th highest	Renault	13.5% (200) [9.4–18.9]	Williams	5.2% (134) [2.6–10.4]
4th lowest	Ferrari	5.0% (319) [3.1–8.0]	McLaren	3.7% (136) [1.6–8.3]
3rd lowest	Mercedes	3.8% (320) [2.2–6.4]	Red Bull	3.7% (136) [1.6–8.3]
2nd lowest	Alfa Romeo	3.3% (120) [1.3–8.3]	Aston Martin	3.0% (133) [1.2–7.5]
Lowest	Aston Martin	2.3% (44) [0.4–11.8]	Sauber	2.1% (48) [0.4–10.9]

Change between eras (same constructorId): McLaren improved most (13.5% → 3.7%), followed by Red Bull (10.7% → 3.7%); Alfa Romeo worsened (3.3% → 12.5%); Mercedes and Ferrari stayed low (about 4–5%). In 2014–2021 the Mercedes and Ferrari intervals do not overlap with Toro Rosso's or McLaren's. **Sensitivity:** counting the 88 ambiguous retirements as mechanical gives 11.2% and 7.0% overall and leaves the top five of 2014–2021 and the top two of 2022–2024 unchanged.

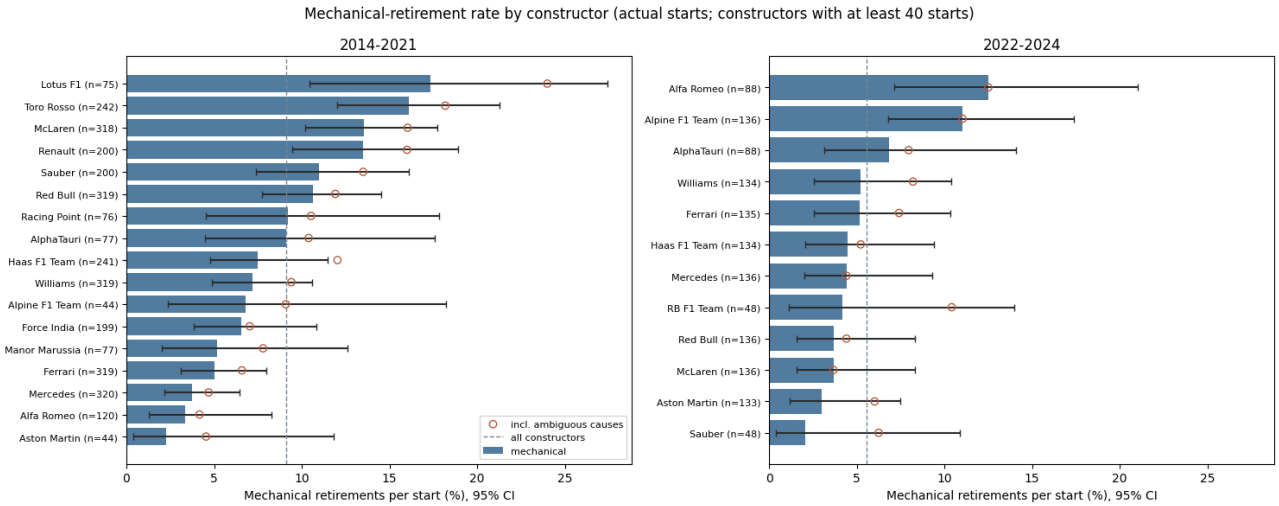


Figure 3. Q3. Mechanical retirements per start by constructor with 95% intervals (bars), the rate if ambiguous causes are counted as mechanical (hollow circles) and the all-constructor average (dashed line).

Limitations. Constructor identity follows constructorId, so a renamed team starts a new history; read by team lineage, the Enstone team (Lotus F1 → Renault → Alpine) is among the highest in both eras. Caterham (23.5% from 34 starts) and Marussia (2014) fall below the ranking threshold. 2022–2024 rates rest on 88–136 starts per team, so the intervals of the top two overlap with several mid-field teams. The status text is the only record of cause.

> **Grain.** one row per driver start (raceId, driverId), 2014–2024; the denominator is actual starts.
> **Missingness.** every status from 2014 onward is mapped (an unmapped status stops the notebook); the 88 ambiguous causes are reported separately and tested as a sensitivity.
> **Pre-race cutoff.** race-ending status is post-race information. It is used only to answer this question and is never a model input.

2. Selected features and why each is valid at the pre-race cutoff

The model table holds actual starters from 2003 onward (8,921 rows; train 2003–2019, validation 2020–2021, test 2022–2024), because qualifying covers every entry only from 2003 (limitation 1). Every feature is recorded in a feature register with its group, source table and the reason it is known before the race; the register is the feature list, so nothing enters the model without a written reason.

Table 6. The 22 model features in 6 groups (Spearman correlation with podium on training rows).

Group	Feature	Definition	Why it is known before the race	Spearman
Grid	grid_position	Starting slot after penalties; pit-lane starts placed behind the field	The grid is final before the start	−0.50
	pit_lane_start	1 if the driver starts from the pit lane	Decided before the start	−0.02
	field_size	Number of cars taking the start	The grid is known before the start	−0.03
Qualifying	q1_gap_pct	% slower than the race's fastest Q1 lap (every driver runs Q1)	Qualifying ends before the race	−0.40
	q1_gap_to_teammate_pct	% slower than the teammate in Q1 (same car)	Qualifying ends before the race	−0.07
	grid_minus_quali	Places lost (penalties) or gained between qualifying and grid	Both fixed before the start	+0.09
Driver form	driver_podium_rate_last5	Share of the previous 5 starts on the podium	Earlier races only: shift(1)	+0.53
	driver_avg_finish_rel_last5	Mean finishing position ÷ field size, previous 5 starts	Earlier races only	−0.44
	driver_finish_rate_last5	Share of the previous 5 starts finished	Earlier races only	+0.16
	driver_form_races	Number of earlier starts behind the rates (0–5)	Earlier races only	+0.08
	driver_experience	log(1 + number of earlier starts)	Earlier races only	+0.23
Constructor form	constructor_podium_rate_last5	Team podiums ÷ team cars, previous 5 races	Team aggregated per race, then shifted	+0.52
	constructor_best_finish_rel_last5	Mean of the team's best relative finish, previous 5 races	Earlier races only	−0.47
	constructor_finish_rate_last5	Share of team cars finishing, previous 5 races	Earlier races only	+0.20
	constructor_form_races	Number of earlier team races (0–5)	Earlier races only	+0.04
Standings	driver_champ_rank	Championship position after the previous race	Snapshot strictly before the race	−0.47
	driver_points_share	Points ÷ leader's points before the race	Snapshot strictly before the race	+0.47
	constructor_champ_rank	Constructors' position after the previous race	Snapshot strictly before the race	−0.45
	constructor_points_share	Team points ÷ leading team's points	Snapshot strictly before the race	+0.48
	season_progress	Share of the season's races already run	The calendar is known in advance	0.00
	standings_from_last_season	1 at season openers (last season's final standings used)	The calendar is known in advance	+0.02
Context	home_race	Driver nationality matches the circuit's country	Fixed before the race weekend	−0.01

Dropped as exact formulas of kept features: grid_relative (= grid_position ÷ field_size) and quali_position (= grid_position − grid_minus_quali); both identities are asserted, so no information is lost. **Not used:** sprint results are

pre-race but exist only from 2021 and would be missing for every training row; in 2021–2022 the sprint set the Sunday grid, so on those 119 validation and test rows grid_minus_quali measures sprint places gained or lost rather than penalties (limitation 6). **Never used:** the current race's results, points, status, laps, pit stops, lap times, fastest lap and post-race standings; a check stops the notebook if any of them enters the feature list.

How leakage is prevented and tested

- **Form** uses `shift(1)` before a rolling 5-race window within each driver, so the current race never enters its own feature. Constructor results are aggregated per team and race *before* shifting, so a teammate's result in the same race cannot leak.
- **Standings.** The standings row of a race holds the standings *after* that race, so each row takes the latest snapshot of a strictly earlier race (`merge_asof` with exact matches excluded). Season openers use the previous season's final standings; rookies and new teams at an opener have no rank (filled later, with a missing flag). Points are expressed as a share of the leader's because the scoring system changed in 2010 (a win went from 10 to 25 points).
- **Perturbation tests** (three for form, three for standings) change one race's result or standings in a copy of the history: the features of that race and of every earlier race stay identical; only later races change.
- **Every missing value has a documented, asserted reason** (for example, a driver's first start has no form). Training missingness is at most 3.6% per feature.

> **Data quirk: McLaren 2007.** McLaren was excluded from the 2007 constructors' championship. The dataset keeps their 2007 points but ranks them last (11th) in every 2007 race, so their constructor championship rank for 2007, and for the 2008 opener (which uses the 2007 final standings), shows the season's strongest team as the weakest: 34 training rows, 23 of them podiums. Constructor points share, driver standings and constructor form are unaffected. The values are kept as recorded; the rows are 0.5% of training and none are in validation or test (limitation 10).

3. Known limitations

1. **Qualifying-coverage drift.** Qualifying covers every entry only from 2003 (1996–2002: 6–58%; none before 1994), so the model is trained on 2003–2019 only. Findings for earlier eras come from the descriptive questions, not the model.
2. **Era sensitivity.** Test PR-AUC drops to 0.651 in 2022, when new technical rules reshuffled the order, against 0.773 in 2023. Form and standings carried over from an old rule set mislead the model until new results build up.
3. **Small validation set.** 39 races, including the shortened 2020 season; many differences between settings are inside the race-bootstrap intervals and are treated as ties.
4. **Threshold drift.** The validation threshold gives lower precision on test (0.689 → 0.617).
5. **The cohort uses post-race fields for scope.** Status and laps decide who took the start. They define which rows are predicted, never a feature, but a driver who withdraws after qualifying would still be scored in real use.
6. **Sprint weekends 2021–2022.** The sprint set the grid, so grid_minus_quali measures sprint places gained or lost rather than penalties on 119 validation and test rows.
7. **Constructor identity.** Teams are followed by constructorId, so a renamed team (Toro Rosso → AlphaTauri → RB) starts a new form history; this also affects the Question 3 rankings.
8. **Historical attribution.** Shared-drive groups decided by the lowest resultId are not independently verified. All are before 1982 and outside the model table.
9. **Explanations are not causal.** Permutation importance, SHAP and LIME show what the model relies on, not what causes a podium.
10. **Retroactive standings.** The dataset applies McLaren's 2007 constructors' exclusion to every 2007 race by ranking them last (their points are kept), so their constructor championship rank for 2007 and the 2008 opener (34 training rows) does not match what was known before those races. Points share, driver standings and form features are unaffected.

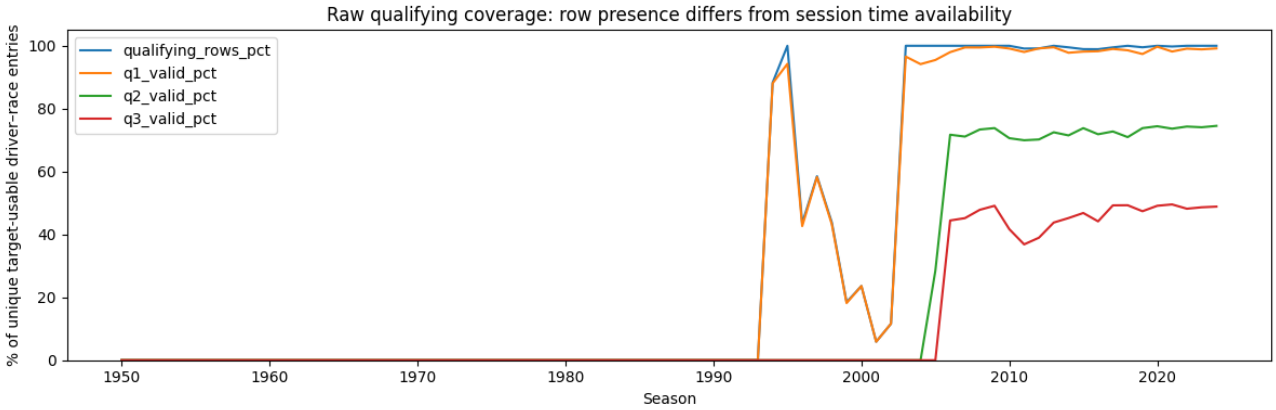


Figure 4. Qualifying coverage per season: share of driver–race entries with a qualifying row and with a valid Q1, Q2 and Q3 time. Coverage is complete only from 2003 (limitation 1).

Table 7. Era sensitivity: final model (logistic regression) per test season (limitation 2).

Test season	ROC-AUC	PR-AUC	F1	Context
2022	0.946	0.651	0.658	New technical rules reshuffled the competitive order
2023	0.931	0.773	0.704	One dominant team: most predictable
2024	0.929	0.666	0.647	Most open season of the three

Known limitations (continued)

11. **The label is the official result.** positionOrder is the classification after stewards' decisions, so a driver disqualified after finishing in the top three counts as no podium (for example Ricciardo, 2014 Australian GP; Hamilton, 2023 United States GP; Russell, 2024 Belgian GP, the last two in the test set). The model predicts the official podium, not the order at the finish line.
12. **Q1 means different things before and after 2006.** In 2003–2004 ranking drivers by their Q1 time reproduces the final qualifying order exactly (Spearman 1.00; 0.83–0.91 from 2006): qualifying was a single timed lap run on race fuel, so q1_gap_pct partly reflects fuel strategy in the first training seasons. From 2006 Q1 is the first low-fuel knockout round.
13. **Optimistic validation scores.** Validation chose the hyperparameters, early stopping, decision thresholds and the final model, so validation scores are slightly optimistic; the test set, scored once after every choice was frozen, is the fair estimate.

4. Model results and explanations

All models use the same 8,921 rows and 22 features. Hyperparameters, early stopping and decision thresholds were chosen on validation (2020–2021); the test seasons (2022–2024) were scored once, after every choice was frozen. **PR-AUC is the primary metric** because podiums are a 15% minority and PR-AUC measures how well true podium finishers are ranked on top; ROC-AUC is secondary. F1, precision and recall are taken at each model's validation-chosen threshold.

Table 8. Model comparison (best attempt per family; 11 attempts in the notebook).

Model	Train PR-AUC	Val PR-AUC	Test PR-AUC	Test ROC-AUC	Test F1	Test precision	Test recall
Logistic regression (final)	0.696	0.723	**0.697**	0.936	0.667	0.617	0.725
Gradient boosting	0.779	0.695	0.709	0.937	0.658	0.665	0.652
Shallow neural network (1×16)	0.708	0.715	0.680	0.931	0.675	0.611	0.755
Rule: podium if grid ≤ 3	0.629	0.639	0.647	0.913	0.649	0.650	0.647

Logistic regression was chosen on validation: neither challenger beat it beyond the race-bootstrap interval, and it is the simplest model. It does not overfit (train 0.696, test 0.697) and beats the grid rule on ranking (test PR-AUC +0.050, 95% interval [+0.004, +0.097]), although the rule is as good at picking each race's top three (65.2% vs 63.7% of picks on the podium). **Ablation:** removing the grid group costs 0.087 validation PR-AUC (this also removes the qualifying position, which was dropped as a formula of the grid); every other group alone costs less than 0.01 because the groups overlap, and together they add about 0.08.

Global explanations. Permutation importance (validation, drop in PR-AUC over 20 shuffles) and SHAP agree: the model relies mostly on starting position (shuffling grid_position costs 0.32 PR-AUC; its mean |SHAP| of 0.085 is more than three times that of the next feature), followed by recent driver and team results (average relative finish, podium and finish rates) and the Q1 gap. Shuffled as whole groups, grid matters most (0.35), then driver form (0.08), standings and qualifying (0.04 each). Standings add little once form is known, consistent with the ablation. Correlated features (grid and Q1 gap, points share and rank) share importance between them.

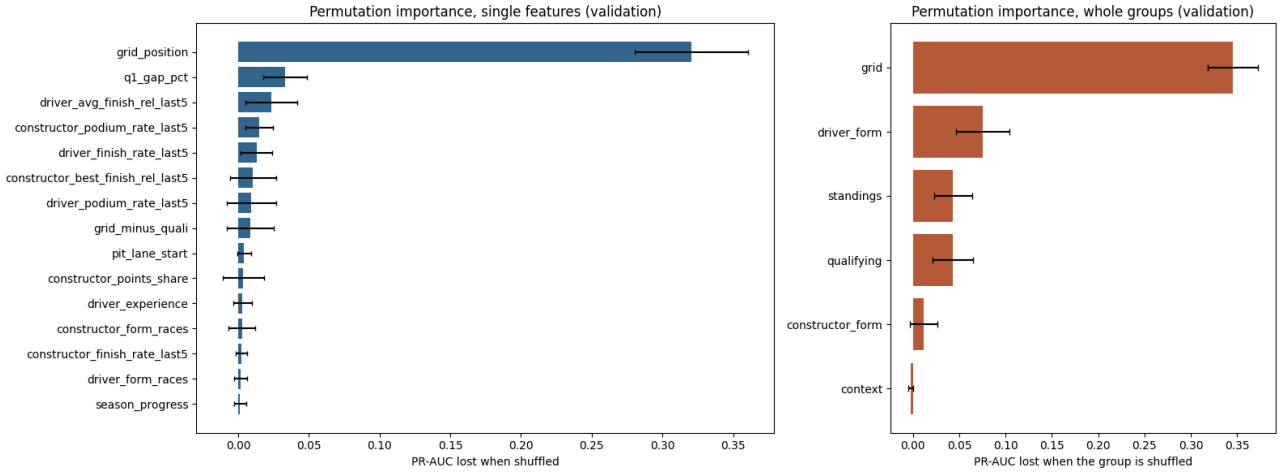


Figure 5. Permutation importance on validation: drop in PR-AUC when a single feature (left) or a whole feature group (right) is shuffled; bars show the mean and standard deviation over 20 shuffles.

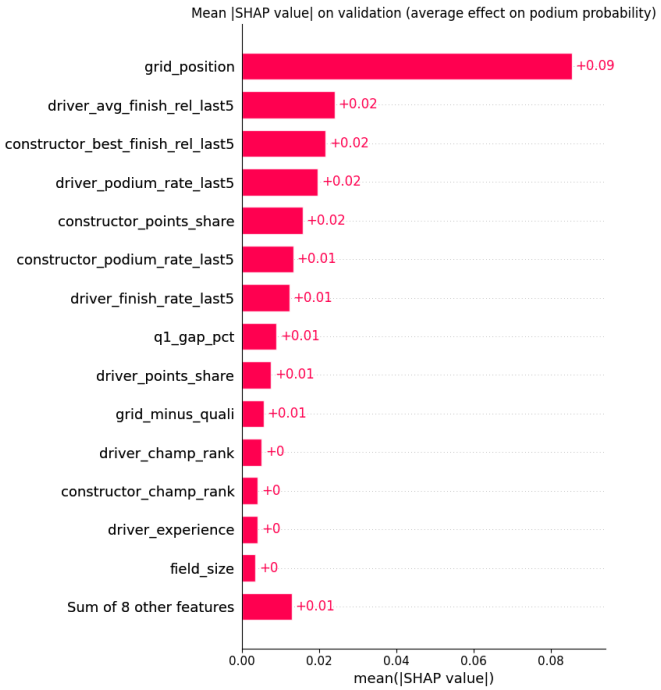


Figure 6. Mean |SHAP value| per feature on validation: the average size of each feature's contribution to the predicted podium probability, relative to the average prediction. Grid position dominates; the next features are recent driver and team results.

Local explanation: Max Verstappen, 2023 Belgian Grand Prix (test set). He set the fastest qualifying time but took a five-place grid penalty and started 6th; the grid rule predicts no podium. The model gave 0.78 (threshold 0.36) and he won. *SHAP:* from the average prediction of 0.10, grid position 6 adds +0.18, his average relative finish over the last five races +0.14, podiums in all of his last five starts +0.11, starting five places behind his qualifying position +0.11 and his team's best relative finish +0.09. *LIME* agrees: grid ≤ 6 (+0.19), starting behind his qualifying position (+0.08) and recent driver and team form. The model recognises a fast car starting out of position, the pattern found in Question 1.

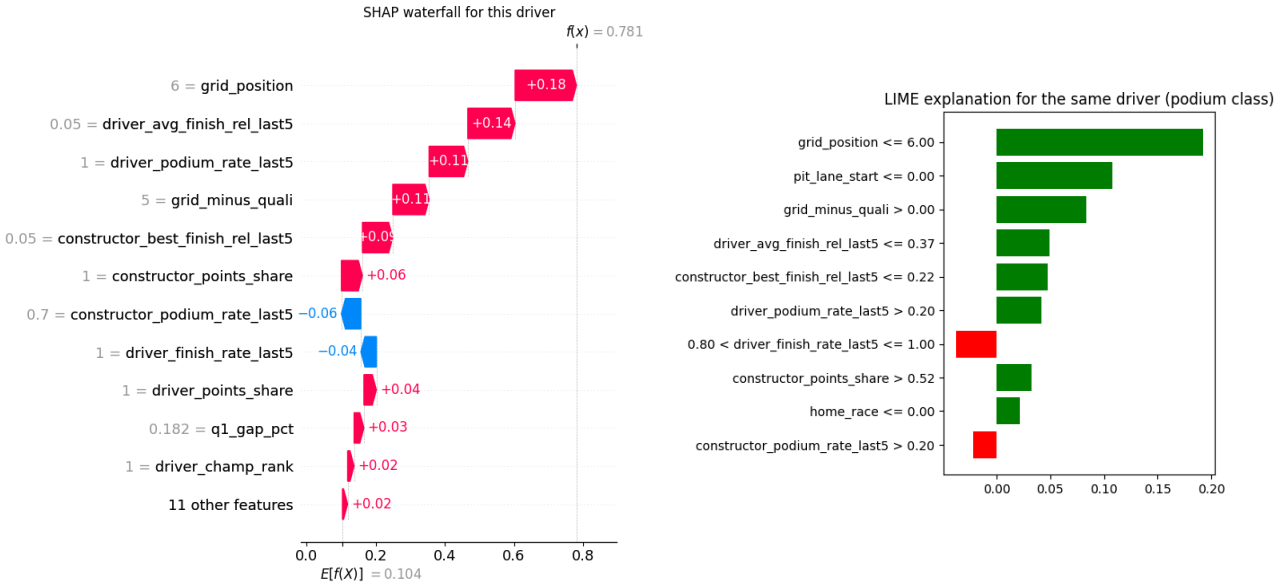


Figure 7. Local explanation of one prediction. Left: SHAP waterfall (contributions add up from the average prediction, 0.10, to this driver's 0.78). Right: LIME weights towards the podium class.

Explanations are not causal. Permutation importance, SHAP and LIME show what the model relies on to make its predictions, not what causes a podium. A good grid slot predicts a podium because fast cars both qualify and finish at the front; it does not show that starting further forward would by itself produce a podium.